

GrounUp Enterprise Specification

Phase 25 - Master Library, Service Catalog, Resource/Material/Equipment, Assembly & Rate Management

Version 1.0.0 | Implementation-ready specification

1. Mission

Implement the governed master-library system that becomes the single reusable source for GrounUp services, resources, assemblies, production rates, price/rate records, regional profiles, tenant overrides and estimator snapshots.

2. Phase 24 Foundation

Phase 25 inherits Phase 24 identity, tenant isolation, authorization, audit, workflow, API/event, configuration and observability controls rather than duplicating them.

3. Service Catalog

Authorized tenant users can create and version services and categories. Approved estimator-created services can write back through the same governed catalog workflow so estimate knowledge becomes reusable enterprise data.

4. Resource Libraries

Labor, crews, equipment, trucks, materials, subcontract resources and disposal records share common identifiers, units, provenance, version and approval metadata while retaining domain-specific fields.

5. Assemblies

Assemblies reference governed components and can nest approved assemblies. Dependency validation prevents missing, incompatible or cyclic references.

6. Rates and Prices

Labor, equipment, material, truck, subcontract and disposal prices retain unit, currency, effective date, geography/profile, source and approval evidence. Equipment supports multiple period rate structures rather than assuming linear conversion.

7. Regionalization and Overrides

Base libraries may be layered with approved regional profiles, vendor/custom prices and tenant overrides according to explicit precedence rules. Double-adjustment is prevented by policy.

8. AI Writeback

AI may propose library additions or updates using evidence and confidence. AI cannot publish; authorized human review may convert an accepted proposal into a governed draft/version.

9. Estimator Reproducibility

The estimator consumes an immutable library snapshot or explicit record versions. Later library updates cannot silently change historical estimates.

10. Implementation Order

Common record/version model -> UOM/conversions -> service/resource libraries -> assemblies -> rates/prices -> regionalization/overrides -> publication resolver -> snapshots -> imports -> AI proposals -> estimator gateway.

Implementation Modules

ID	Module	Layer	Purpose
P25-MOD-001	Library Registry	Core	Global registry of library types, versions, ownership and publication state
P25-MOD-002	Service Catalog Service	Domain	Services, categories, units, estimating mappings and downstream references
P25-MOD-003	Unit of Measure Service	Core	Canonical units, conversions and dimensional validation
P25-MOD-004	Labor Resource Service	Domain	Craft/classification, burden inputs and labor resource definitions
P25-MOD-005	Crew Template Service	Domain	Crew compositions, default sizes and production references
P25-MOD-006	Equipment Resource Service	Domain	Equipment classes/models/capacity and owning/operating rate references
P25-MOD-007	Truck Resource Service	Domain	Truck classes, capacity, hourly/trip references and haul configuration
P25-MOD-008	Material Resource Service	Domain	Material definitions, units, waste factors, density and source references
P25-MOD-009	Subcontract Resource Service	Domain	Subcontract categories, unit rates, vendor/source references
P25-MOD-010	Disposal Service	Domain	Disposal facilities/categories, fees, units, restrictions and source references
P25-MOD-011	Assembly Service	Domain	Reusable estimate assemblies and component dependency graph
P25-MOD-012	Production Rate Service	Domain	Task/equipment/crew production rates with conditions and provenance
P25-MOD-013	Labor Rate Service	Pricing	Base wage, burden, union/custom profile and effective dates
P25-MOD-014	Equipment Rate Service	Pricing	Hourly/daily/weekly/monthly rate structures and fuel/operating inputs
P25-MOD-015	Material Price Service	Pricing	Supplier/material prices, freight references and effective dates
P25-MOD-016	Truck Rate Service	Pricing	Hourly, load, trip and haul-rate definitions
P25-MOD-017	Subcontract Rate Service	Pricing	Vendor/subcontract unit-price references
P25-MOD-018	Regional Profile Service	Pricing	Location factors, geography hierarchy and price-profile selection
P25-MOD-019	Vendor Price Source Service	Pricing	Quote/source documents, expiry and approved vendor price references
P25-MOD-020	Tenant Override Service	Configuration	Tenant-specific editable overrides without mutating base records
P25-MOD-021	Publication & Version Service	Governance	Draft/review/approved/published/superseded lifecycle
P25-MOD-022	Import/Bulk Edit Service	Operations	CSV/JSON imports, mapping, validation, preview and governed commit
P25-MOD-023	AI Library Proposal Service	AI	AI-proposed additions/updates with source evidence and human approval
P25-MOD-024	Estimator Library Gateway	Integration	Stable read/query/snapshot contracts for Phase 26 estimator implementation